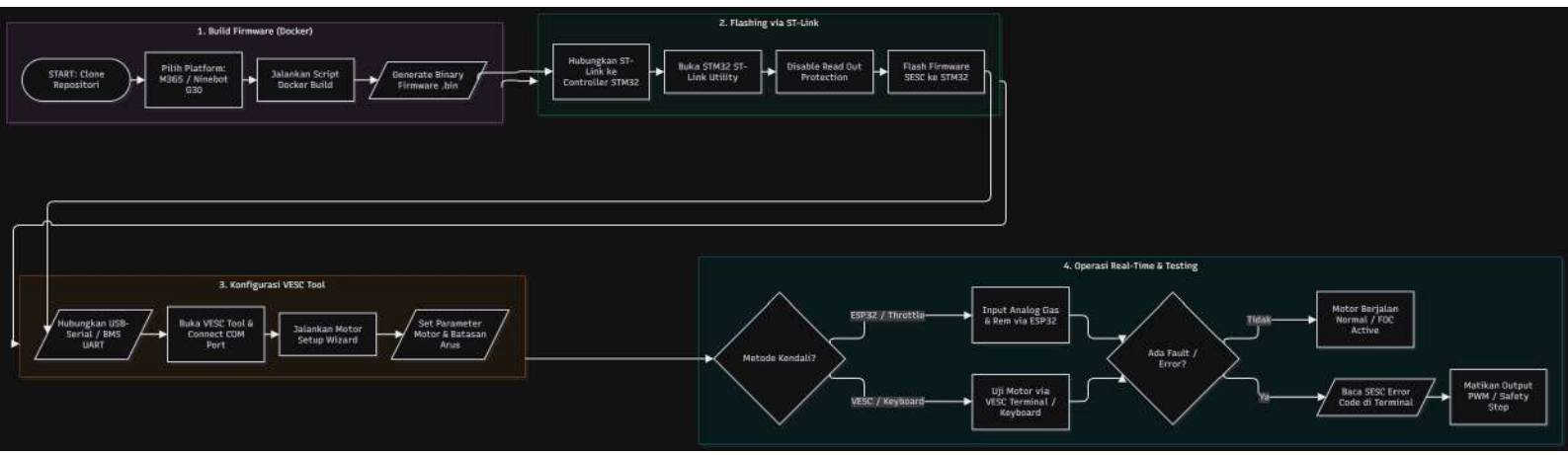
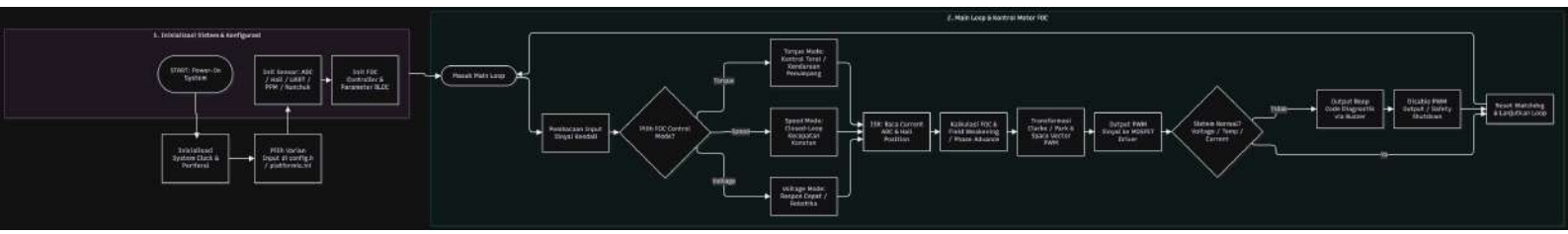

1. Inisialisasi Sistem
Start: Power-On
Init Clock & Interrupt
Init Peripherals: GPIO, ADC, UART
Baca Parameter EEPROM
Init Encoder & Hall Sensor
Init FOC & VESC Module
Main Loop
2. Main Loop & Interrupt Services
Ada Input UART?
Ya
Tidak
Menerima Data UART
Parsing Protokol VESC
Update Target Setpoint
Kirim Status Telemetri
Trigger Timer & ADC Interrupt
Baca Sensor: ADC Current & Encoder
Kalkulasi FOC & Clarke/Park
Generate Vector Space PWM
Output Sinyal PWM ke MOSFET
Sistem Aman?
Tidak
Ya
Trigger Fault: Matikan PWM
Reset Watchdog Timer


1. Build Firmware (Docker)
START: Clone Repositori
Pilih Platform: M365 / Ninebot G30
Jalankan Script Docker Build
Generate Binary Firmware .bin
2. Flashing via ST-Link
Hubungkan ST-Link ke Controller STM32
Buka STM32 ST-Link Utility
Disable Read Out Protection
Flash Firmware SESC ke STM32
3. Konfigurasi VESC Tool
Hubungkan USB-Serial / BMS UART
Buka VESC Tool & Connect COM Port
Jalankan Motor Setup Wizard
Set Parameter Motor & Batasan Arus
4. Operasi Real-Time & Testing
Metode Kendali?
ESP32 / Throttle
VESC / Keyboard
Input Analog Gas & Rem via ESP32
Uji Motor via VESC Terminal / Keyboard
Ada Fault / Error?
Tidak
Ya
Motor Berjalan Normal / FOC Active
Baca SESC Error Code di Terminal
Matikan Output PWM / Safety Stop


1. Inisialisasi Sistem & Konfigurasi
START: Power-On System
Init Sensor: ADC / Hall / UART / PPM / Nunchuk
Init FOC Controller & Parameter BLDC
Inisialisasi System Clock & Peripheral
Pilih Varian Input di config.h / platformio.ini
Masuk Main Loop
2. Main Loop & Kontrol Motor FOC
Pembacaan Input Sinyal Gasball
Pilih FOC Control Mode?
Torque
Speed
Voltage
Torque Mode: Kontrol Torsi / Kendaraan Penumpang
Speed Mode: Closed-Loop Kecepatan Konstan
Voltage Mode: Sangat Cepat / Robotika
FOC: Baca Current ADC & Hall Position
Kalkulasi FOC & Field Weakening / Phase Advance
Transformasi Clarke / Park & Space Vector PWM
Output PWM Sinyal ke MOSFET Driver
Sistem Normal? Voltage / Temp / Current
Tidak
Ya
Output Beep Code Diagnostik via Buzzer
Disable PWM Output / Safety Shutdown
Reset Watchdog & Lanjutkan Loop